

Science Olympiad — Final Challenge Round 3

Names of participants: (Please print neatly)

School Name:

Team Number:

Check your team: ☐ Varsity ☐ JV1 ☐ JV2 ☐ JV3

Warning: Do not open this packet until given permission to do so.

Note: There are useful notes after this page.

Question	Value	Incorrect letters	Deduction	Score
1	0			
2	0			
3	0			
4	0		≡	
Final Score				

1) **[0 points]** The following quote needs to be decoded with the Aristocrat Cipher using a K1 alphabet. What does it say?

NPR NM GYR MUGYRST TZXPQR YZT PUDR ZP WNDZWUCCK

CUSXR MUTYZNP UPQ CRMG NGYRST IZGY CZGGCR GN PN SNND

K1	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
Frequency			6	3			8		1		1		4	9		7	2	9	4	5	6		2	2	6	7
Replacement																										

2) **[0 points]** Solve this quote that has been encoded using the Baconian Cipher. You are told that the deciphered text starts with **I ORD**. What does it say?

A HOME ALVIN BOOK A SOUTH WIVES DECAY IS NOW WASN'T

WORST JIMMY STRAP A PART BEEPS GARTH AS HIS EMAIL

AT YOU CELLS COMET EERIE ARABS CHINA GEESE ENJOY

MOUTH EATIN KARMA CLIVE TOURS MAJOR NECKS IS TOO

MILES

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
Replacement																										

3) [0 points] Your 3D printer paused mid-print to extrude this mysterious Homophonic string using a keyword that has the letters **M** and **T** in it. What does it say?

89 64 96 54 6 87 00 40 83 41 99 47 95 76 39 94 15 60 19 50

55 49 87 63 76 17 58 55 19 7 87 39 76 87 47 92 00

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25
26	27	28	29	30	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45	46	47	48	49	50
51	52	53	54	55	56	57	58	59	60	61	62	63	64	65	66	67	68	69	70	71	72	73	74	75
76	77	78	79	80	81	82	83	84	85	86	87	88	89	90	91	92	93	94	95	96	97	98	99	00

4) [0 points] Solve this

Values to decode for solution

No known solution mapping has been generated. Please click the Check Problem and Update button

Cryptarithm formula

0 1 2 3 4 5 6 7 8 9